Table 1: Signed change in metabolic tumor volume (MTV) and total lesion glycolysis (TLG) caused by registration on the test set (26 patients, 30 lesion-positive pairs): positive values mean that the warped lesions are larger or have greater total uptake. Methods are as in Table **??**; Training / IO indicates where losses are used, and PET + rig. denotes PET-preservation and bone-rigidity losses. Values are mean±SD and median over patients after averaging each patient's eligible pairs.

			ΔMTV (%)		ΔTLG (%)	
Method	Training	IO	Mean±SD	Median	Mean±SD	Median
Affine	–	–	$+2.9 \pm 5.8$	+1.3	$+2.8 \pm 5.7$	+1.7
NiftyReg	–	–	$+4.7 \pm 18.9$	−0.6	$+7.1 \pm 23.8$	−0.1
ConvexAdam	–	–	$+8.0 \pm 8.1$	+7.1	$+8.8 \pm 10.3$	+5.8
Affine + IO	–	PET preservation + bone rigidity	$+3.5 \pm 5.8$	+3.8	$+5.7 \pm 5.6$	+6.7
Ours	–	–	$+5.3 \pm 13.1$	+4.5	$+7.9 \pm 14.9$	+7.0
Ours	PET preservation + bone rigidity	–	$+5.1 \pm 11.0$	+3.4	$+7.6 \pm 12.5$	+5.2
Ours + IO	–	basic	$+1.1 \pm 16.5$	+2.1	$+4.0 \pm 20.0$	+3.6
Ours + IO	PET preservation + bone rigidity	PET preservation + bone rigidity	$+4.2 \pm 11.8$	+3.0	$+7.3 \pm 12.8$	+6.4
Ours + IO	–	PET preservation + bone rigidity	$+3.2 \pm 13.5$	+3.1	$+6.2 \pm 14.8$	+6.2